

# The Substrate Framework: Axioms, Constraints, and Emergent Structure

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## Abstract

This document presents a high-level overview of the Substrate Framework, a universal-Hilbert-space ontology in which the physical world emerges from unitary dynamics and information-theoretic constraints. The framework is built on three axioms (Hilbert Space Realism, Unitary Evolution, and Emergent Classicality) and several consistency constraints (No-Signaling, Finite Effective Information Density, Stability of Emergent Structures, and Quantum Consistency). The goal is to provide a minimal, coherent foundation from which particles, spacetime, interactions, and classical dynamics arise as emergent, self-organizing patterns within a single universal wavefunction.

## 1 Axiom 1: Hilbert Space Realism

**Axiom 1.** *The fundamental ontology of the world is a universal Hilbert space  $\mathcal{H}$  and its state vector  $|\Psi\rangle \in \mathcal{H}$ .*

The Hilbert space is not a mathematical convenience; it is the arena of existence. Systems, particles, fields, observers, measurement devices, and spacetime structures arise as dynamically persistent submanifolds, patterns, and factorizations within  $|\Psi\rangle$ . There are no fundamental particles or fields living on spacetime. Instead, “physical” structures are emergent, stable, decohered subspaces embedded in the universal state.

## 2 Axiom 2: Unitary Evolution

**Axiom 2.** *The evolution of the universal state is unitary.*

On all scales and at all times, the dynamics of  $|\Psi(t)\rangle$  are governed by unitary evolution:

$$|\Psi(t + \Delta t)\rangle = U(\Delta t) |\Psi(t)\rangle, \quad (1)$$

for some unitary operator  $U$  or, in continuous time,

$$i\hbar \frac{d}{dt} |\Psi(t)\rangle = H |\Psi(t)\rangle, \quad (2)$$

for Hermitian  $H$ . No collapse postulate is introduced; all apparent collapses or measurements are effective descriptions of branching and decoherence inside  $\mathcal{H}$ .

### 3 Axiom 3: Emergent Classicality

**Axiom 3.** *Classical reality emerges from stable, decohered structures within the universal wave-function.*

Macroscopic definiteness (trajectories, positions, tables, detectors, and classical fields) arises from the internal organization of  $|\Psi\rangle$ . Classical states correspond to:

- approximately decohered sectors,
- dynamically stable patterns,
- predictable coarse-grained variables.

This axiom does not add new dynamical rules; it asserts that classical physics is the large-scale organizational limit of the Hilbert-dynamical substrate.

### 4 Constraint 1: No-Signaling and Emergent Spacetime

**Constraint.** *Observers must not be able to send controllable superluminal signals.*

This operational requirement imposes a structural condition on the emergent description: locality and causal cones must arise as coarse-grained constraints. The no-signaling condition forces:

- an emergent causal order,
- finite-speed propagation of influences,
- effective Lorentz-like kinematics in the low-energy regime.

Thus, spacetime is not fundamental but is a geometric encoding of causal constraints that emerge from allowed information flows in  $\mathcal{H}$ .

### 5 Constraint 2: Finite Effective Information Density

**Constraint.** *Emergent regions of spacetime must have finite effective information capacity.*

Although  $\mathcal{H}$  may be unbounded, any emergent classical region must behave as if it has limited information storage. This requirement motivates:

- localized “substrate cells” or modes,
- constraints on entanglement spread,
- effective geometric structures reminiscent of area or volume laws.

This constraint underlies the “defrag” mechanism, which collects fragmented substrate information into localized stable lumps (proto-particles) and drives the emergence of gravitational-like organization.

## 6 Constraint 3: Stability of Emergent Structures

**Constraint.** *Some patterns in the universal state must be dynamically stable and re-identifiable over time.*

Effective particles, fields, and classical objects exist only if the dynamics produce persistent, self-maintaining clusters of entanglement and information. These emergent structures must:

- survive environmental interaction,
- preserve identity under time evolution,
- generate effective interaction laws.

This constraint is functional: without stable patterns, no classical world exists.

## 7 Constraint 4: Quantum Consistency

**Constraint.** *Emergent dynamics must be consistent with known quantum limits.*

The framework must reproduce:

- Bell and CHSH inequalities up to the Tsirelson bound,
- the Born rule at the observational level,
- approximate unitarity and locality for effective subsystems,
- entanglement-entropy scalings compatible with quantum field theory.

Any proposed substrate Hamiltonian or rule set must be tested against these requirements.

## 8 Scope and Purpose

The Substrate Framework is not a specific model but a set of principles governing which models are admissible. Its purpose is:

- to provide a minimal ontology (Hilbert space only),
- to derive classical and relativistic physics as emergent organizational phenomena,
- to explain particle-like excitations as dynamically stable information clusters,
- to unify quantum dynamics and emergent geometry under one structure.

The framework is compatible with a wide variety of concrete substrate models, including lattice Hamiltonians, tensor networks, dynamical graph substrates, and continuum Hilbert-field models. The goal is to identify which of these families realize the axioms and constraints in a self-consistent way.

## 9 Outlook

Future work aims to:

- derive specific particle-like excitations from substrate dynamics,
- demonstrate that causal structure and finite influence speed emerge naturally,
- connect the “defrag” potential to gravitational behavior,
- explore how fermionic statistics and gauge fields emerge as stability conditions.

These directions collectively aim toward a unification of quantum theory, classical mechanics, and relativistic geometry within a single Hilbert-space substrate.